

Sebastian G. Gruber

Machine Learning Engineer & Data Scientist

PhD in Machine Learning | MSc in Statistics

Frankfurt am Main, Germany

[linkedin.com/in/sebastian-gruber-21b76813b](https://www.linkedin.com/in/sebastian-gruber-21b76813b) · sebggruber.github.io · [Google Scholar](#)



PROFILE

Machine learning engineer and statistician by training (BSc/MSc Statistics, PhD in machine learning, summa cum laude) who builds data-driven systems end to end – from data generation, validation and statistical analysis to fine-tuned models, cloud pipelines and product integration. Currently at LINK2AI, taking an LLM hallucination detector from first experiments to a GPU-served module of the company's LLM monitoring platform. Particular strengths: uncertainty quantification and calibration, rigorous evaluation, and understanding how the data-generating process shapes what a model learns. First-author papers at NeurIPS, ICML, AISTATS and UAI; 300+ citations.

SKILLS

Statistics: statistical modelling & inference, regression, bootstrap confidence intervals, hypothesis testing, confounder analysis, uncertainty quantification & calibration, proper scoring rules, inter-annotator agreement

Machine learning: supervised learning (e.g. random forests), deep learning, fine-tuning of LLMs and encoders (LoRA/QLoRA, 4-bit quantisation), LLM evaluation & LLM-as-a-judge, RAG, embeddings, synthetic data generation

Programming: Python (NumPy, pandas, SciPy, scikit-learn, PyTorch, Hugging Face Transformers/PEFT, vLLM), R (mlr3), SQL, Julia

MLOps & engineering: AWS, Terraform, Docker, Ray Serve, MLflow, Optuna, GitLab CI, FastAPI, Streamlit, Linux, Git, pytest; OpenAI/Anthropic APIs, LiteLLM; AI-assisted development (Claude Code)

EXPERIENCE

Machine Learning Engineer

LINK2AI GmbH

Feb 2026 – present

Frankfurt am Main

Start-up for the security and trustworthiness of LLM applications (runtime monitoring and protection platform).

- **Hallucination detection for RAG:** owned the answer-grounding sensor end to end within an ESA BIC Hessen incubation project – problem framing, data, evaluation, model selection and product integration – to flag LLM answers that are not supported by the retrieved sources.
- **Evaluation & statistics:** built a reproducible benchmarking framework on public and domain-specific data (per-class error rates, AUROC, latency percentiles, MLflow tracking); used partial correlations, fixed-effects regression and bootstrap confidence intervals to identify what drives performance, showing that how a dataset was constructed – its data-generating process – largely determines what models learn from it.
- **Data:** created synthetic, domain-specific evaluation sets from technical documents, labelled by a multi-vendor LLM-judge panel and validated against blind human annotation (Cohen's κ); uncovered substantial label noise in public benchmarks.
- **Modelling:** fine-tuned and compared ModernBERT encoders and 4-bit LoRA-adapted LLMs (Qwen, Mistral, Gemma) against NLI, embedding, open-source detector and commercial LLM-judge baselines; an on-premise fine-tuned model is competitive with commercial LLM judges at an order of magnitude lower latency, and per-deployment threshold calibration proved the largest single lever.
- **Product integration:** implemented the detector as a GPU-served module of the company's monitoring platform (PyTorch, Ray Serve, 4-bit inference) with structure-aware chunking of long documents, hardened parsing of uploaded PDF/Office/HTML files, a configuration UI (Next.js/React), GPU tests in CI and documentation.
- **Policy classifiers & synthetic data:** built a pipeline that turns a chatbot's compliance policy into one fine-tuned lightweight classifier per rule – LLM-generated training data with diversity control (own Python library: Vendi score and spectral topic extraction on embeddings), Optuna sweeps tracked in MLflow, export to the Hugging Face Hub and serving via vLLM.

Postdoctoral Researcher in Machine Learning

KU Leuven, ESAT-PSI (Prof. Matthew Blaschko)

Apr 2025 – Dec 2025

Leuven, Belgium / remote

- Uncertainty calibration for trustworthy AI, e.g. confidence-aware estimation of medical-imaging biomarkers (MIDL 2026); co-supervised PhD students, resulting in three co-authored papers.
- Co-organised the AISTATS 2026 workshop *Towards Trustworthy Predictions: Calibration for Modern AI* with researchers from Inria, Microsoft Research and UPenn (33 accepted papers); NeurIPS 2025 Top Reviewer.

Doctoral & Postdoctoral Researcher in Machine Learning

German Cancer Research Center (DKFZ), ML in Oncology group (Prof. F. Buettner)

Mar 2021 – Mar 2025

Frankfurt am Main

- Developed a statistical framework for quantifying the uncertainty of ML predictions (proper scores, bias–variance decompositions, calibration-error estimators), from classifiers to generative models and LLMs; PhD summa cum laude based on six peer-reviewed papers.
- Three-month research visit at Inria Paris (Francis Bach), funded by a DKFZ travel grant; resulted in a TMLR paper.
- Supervised eight theses (seven BSc, one MSc); lectures, tutorials and exams for *Introduction to Methods of AI* at Goethe University (2021–2024).
- Nine conference and invited talks, incl. Bayer AG Research (2022) and the Helmholtz AI Conference (2024).

Working Student & Master's Thesis, Machine Learning

Siemens AG

Aug 2018 – Feb 2021

Munich

- **Production ML pipeline** (IT Analytics Lab): built a live prediction service on AWS, provisioned with Terraform, that queries a company database every 10 minutes, scores new records with a random-forest model and writes the predictions back – starting automatically every workday morning and shutting down in the evening to save cloud costs.
- **Research** (Corporate Technology, Machine Intelligence): post-hoc uncertainty calibration under domain shift, resulting in a CVPR 2021 paper (130+ citations) and a US patent application; master's thesis on meta-learning and knowledge distillation.

Working Student, Data Analytics

Criteo

Jul 2017 – Jul 2018

Munich

- Designed and built a web application that automatically generates quarterly business reports.

EDUCATION

PhD in Computer Science (Dr. rer. nat.) – summa cum laude

Goethe University Frankfurt

Mar 2021 – Jan 2025

Frankfurt am Main

Thesis: *A Novel Framework for Uncertainty Quantification via Proper Scores for Classification and Beyond*

MSc in Statistics – grade 1.4

Ludwig Maximilian University (LMU)

Oct 2018 – Feb 2021

Munich

Thesis on meta-learning and knowledge distillation (with Siemens Corporate Technology)

BSc in Statistics – grade 1.43

Ludwig Maximilian University (LMU)

Oct 2015 – Sep 2018

Munich

Thesis on reinforcement learning

SELECTED PUBLICATIONS

- **Gruber, S. G.**, Walha, N., Bach, F., Buettner, F. (2026). *Eigenvalue Calibration for Semantic Embeddings of Large Language Models*. UAI.
- Walha, N., **Gruber, S. G.**, et al. (2026). *Fine-grained Uncertainty Decomposition in Large Language Models: A Spectral Approach*. AAAI.
- **Gruber, S. G.**, Buettner, F. (2024). *A Bias-Variance-Covariance Decomposition of Kernel Scores for Generative Models*. ICML.
- Popordanoska, T. *, **Gruber, S. G.** *, et al. (2024). *Consistent and Asymptotically Unbiased Estimation of Proper Calibration Errors*. AISTATS.
- **Gruber, S. G.**, Buettner, F. (2022). *Better Uncertainty Calibration via Proper Scores for Classification and Beyond*. NeurIPS.
- Tomani, C., **Gruber, S. G.**, et al. (2021). *Post-hoc Uncertainty Calibration for Domain Drift Scenarios*. CVPR (oral).

*equal contribution. 11 papers and one US patent application (17/347,645); 300+ citations – Google Scholar.

ADDITIONAL

Open source: mlr3hyperband (CRAN package for hyperparameter optimisation in mlr3); mlrPlayground (interactive web app teaching ML algorithms)

Awards: PhD summa cum laude; NeurIPS 2025 Top Reviewer; DKFZ short-term travel grant (2024)

Languages: German (native), English (fluent), French (basic)